

FEDERICO COLOMBO

Computer Graphics Engineer • Technical Artist

Specialized in **real-time rendering**, low-level **engine architecture (C++)**, and **procedural pipelines**. Graduated with honors (**110L**) in Cinema & Media Engineering from Politecnico di Torino with active R&D in bio-adaptive volumetric SDF raymarching and mobile XR optimization.

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 github/federicocolombo12



WORK EXPERIENCE

Technical Artist & Producer (XR Master Thesis R&D) 03/2026 - Present
Lemons in the Room • Firenze / Remote

- Architected a real-time bio-adaptive volumetric raymarching framework in HLSL for Meta Quest 3, rendering dynamic Signed Distance Fields (SDFs) driven by BLE biometric telemetry.
- Engineered GPU optimizations (Dynamic Bounding Volumes, adaptive ray stepping) locking **90 FPS on standalone XR hardware** (61.3% GPU time reduction, 0B GC alloc).
- Authored Morten Mikkelsen Surface Gradient shading on procedural volumes to eliminate triplanar UV seams, plus a sub-millisecond GPU Voxelizer for particle swarms.

Lead Technical Artist & Graphic Programmer 11/2024 - 07/2026
LEVEL UP LAB (PoliTo Student Team) • Torino, Italy

- Authored custom shaders in HLSL (retro CRT curvature, post-processing distortions) and programmed core gameplay systems in C# for *Line Operator*.
- Directed multidisciplinary student team of 15+ artists, sound designers, and engineers, enforcing Git LFS versioning and milestone sprints.

Teaching Assistant in 3D Computer Animation 10/2024 - 01/2025
Politecnico di Torino • Torino, Italy

- Mentored 80+ engineering students in 3D animation, skeletal rigging, procedural shading, and rendering pipelines in Maya and Blender.
- Led lab sessions covering linear algebra for 3D transforms, quaternions, lighting equations, and real-time graphics optimizations.

Technical Artist & XR Pipeline Developer 03/2023 - 07/2023
RKH SRL • Torino, Italy

- Built interactive VR/AR client experiences and developed automated Python cleanup scripts for RealityCapture photogrammetry datasets (40% faster turnaround).
- Implemented optimized PBR materials and baked lighting setups for standalone mobile VR.

EDUCATION

M.Sc. in Cinema & Media Engineering 2023 - 2026
Politecnico di Torino • **110/110 con Lode (Honors)**

Focus: Real-Time Rendering, Game Engine Architecture, Computer Vision, Digital Signal Processing. Thesis: *PulseEngine — Volumetric Raymarching & Dynamic SDFs in XR*.

B.Sc. in Cinema & Media Engineering 2020 - 2023
Politecnico di Torino • Torino, Italy

Linear Algebra, Multivariate Calculus, Discrete Mathematics, Algorithms & Data Structures, C/C++ OOP.

CORE COMPETENCIES

Real-Time Rendering & Shaders

Compute shaders, Raymarching & SDFs, PBR equations, Deferred shading, SPOM, HLSL/GLSL.

Engine Architecture (C++)

Custom OpenGL engine development, memory layout, draw-call batching, multi-threading.

XR & Standalone Optimization

Meta Quest 3 (Snapdragon XR2), foveated raymarching, 90 FPS locked latency budgets.

Procedural DCC Pipelines

Blender Python API tooling, Unreal PCG & Niagara, automated asset pipelines.

SOFTWARE & TOOLS

 C++ (17/20, STL)	Advanced
 HLSL / GLSL	Compute/Ray
 Unity 6 (URP, C#)	Expert
 Unreal Engine 5	PCG, Niagara
 Blender & Python API	Add-ons
 Python 3	Pipeline
 RenderDoc & Nsight	Profiling
 Git & Git LFS	CI/VCS

LANGUAGES

Italian	Native Speaker
English	Professional Working (C1)